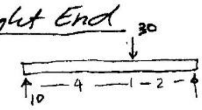
Statics Internal Forces: Shear and Bending Moment Diagrams
(How to Write V and M Equations)

Additional Notes on Writing Equations for V & M

① One important technique is to be able to Write Equations from Right End when one is interested in that end of the beam. Example:

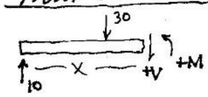


Must remember 2 things:

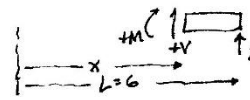
(a) Draw V & M in their (+) sense

(b) Always reference x to left end.

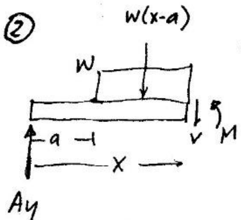
From Left End $(4 \le x \le 6)$



From Right End $(4 \le x \le 6)$



② Uniform Distributed Load



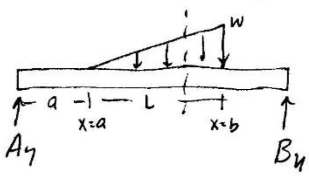
$+\uparrow \sum F_y = 0; \quad A_y - W(x-a) - V = 0; \quad \boxed{V = A_y - W(x-a)}$

$+\circlearrowleft \sum M_s = 0; \quad M - A_y x + W(x-a)\frac{(x-a)}{2} = 0; \quad \boxed{M = A_y x - \frac{W}{2}(x-a)^2}$

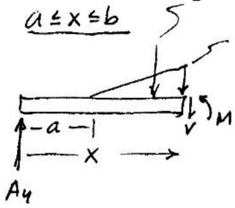
NOTE THAT $M = \int V dx$. A good checker

③ Ascending Triangular Loads

$\frac{1}{2}bh = \frac{1}{2}(x-a)\left[\frac{W}{L}(x-a)\right] = \frac{W}{2L}(x-a)^2$



FBD $a \le x \le b$



$+\uparrow \sum F_y = 0; \quad A_y - V - \frac{W}{2L}(x-a)^2 = 0;$

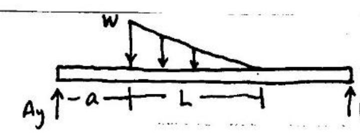
$$\boxed{V = A_y - \frac{W}{2L}(x-a)^2}$$

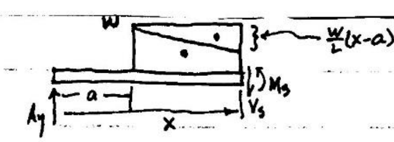
$+\circlearrowleft \sum M_s = 0; \quad M - A_y(x) + \left[\frac{W}{2L}(x-a)^2\right]\left[\frac{(x-a)}{3}\right] = 0$

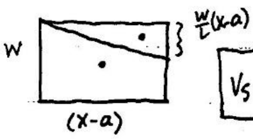
$$\boxed{M = A_y x - \frac{W}{6L}(x-a)^3}$$

④ Descending Triangular Loads

DESCENDING



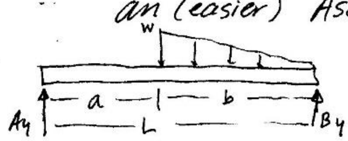




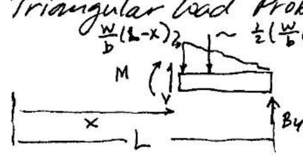
$$\boxed{V_s = A_y - \Big[\underbrace{W(x-a)}_{\text{Area of Rectangle}} - \underbrace{\tfrac{W}{2L}(x-a)^2}_{\text{Subtracting Area of Triangle}}\Big]}$$

$$\boxed{M_s = A_y x - \Big[\underbrace{\tfrac{W}{2}(x-a)^2}_{\text{Moment due to Rectangle}} - \underbrace{\tfrac{W}{6L}(x-a)^3}_{\text{Subtracting Moment due to the triangle.}}\Big]}$$

⑤ Descending Triangular Loads (Writing Equation from Right End makes it an (easier) Ascending Triangular Load Problem)



$\frac{1}{2}\left(\frac{W}{b}(L-x)\right)(L-x) = \frac{W}{2b}(L-x)^2$



$+\uparrow \sum F_y = 0; \quad V - \frac{W}{2b}(L-x)^2 + B_y = 0; \quad \boxed{V = \frac{W}{2b}(L-x)^2 - B_y}$

$+\circlearrowleft \sum M = 0; \quad M + \left[\frac{W}{2b}(L-x)^2\right]\left[\frac{(L-x)}{3}\right] - B_y(L-x) = 0; \quad \boxed{M = B_y(L-x) - \frac{W}{6b}(L-x)^3}$